CLINDAMYCIN

Highest C. difficile risk

Good bone penetration

CHEMICAL NAME Methyl 7-chloro-6,7,8-trideoxy-6-[[(4R)-1-methyl-4-propyl-L-prolyl]amino]-1-thio-L-threo-α-D-galacto-octopyranoside (hydrochloride or phosphate)

CLASS Lincosamide antibiotic

TRADE NAMES Dalacin C · Cleocin · Clindacin · Dalacin T (topical)

THERAPEUTIC USES / INDICATIONS

- Skin and soft-tissue infections (incl. MRSA-susceptible strains), cellulitis in penicillin allergy.
- Anaerobic infections: lung abscess, aspiration pneumonia, dental infections, pelvic infections.
- Bone and joint infections.
- Toxic shock and necrotising fasciitis (adjunct; suppresses toxin production).
- Topical: acne; vaginal: bacterial vaginosis.

ADMINISTRATION

- **PO** capsules with a full glass of water (oesophageal irritation).
- **IV** infusion diluted, max 30 mg/min (≥10–60 min) — never undiluted bolus (cardiac arrest).
- **IM** max 600 mg per site.

DOSAGES

- **Adult PO:** 150–450 mg every 6 h.
- **Adult IV/IM:** 600 mg–2.7 g/day in 2–4 divided doses (severe up to 4.8 g/day).
- **Necrotising fasciitis:** 600–900 mg IV every 8 h.
- **Children:** PO 12–25 mg/kg/day in 4 doses; IV 15–25 mg/kg/day in 3–4 doses (severe 25–40 mg/kg/day).

PHARMACOKINETICS

- **Peak:** PO 45 min. Bioavailability ~90%.
- Excellent bone and abscess penetration; poor CSF.
- Metabolised in liver (CYP3A4); excreted in bile and urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Binds **50S ribosomal subunit** → inhibits protein synthesis (bacteriostatic).
- Active against Gram-positive cocci and anaerobes; inhibits bacterial toxin production.
- No activity against aerobic Gram-negatives.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, abdominal pain.
- Metallic taste (IV).
- Rash.

ADVERSE EFFECTS (SERIOUS)

- *C. difficile*-associated diarrhoea and pseudomembranous colitis (can be fatal, may occur weeks after).
- Severe skin reactions (SJS, DRESS).
- Hepatotoxicity; neutropenia.
- Cardiac arrest with rapid IV injection.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to clindamycin or lincomycin.
- History of *C. difficile* colitis.
- **Caution:** GI disease, elderly, hepatic impairment.

KEY INTERACTIONS

- **Neuromuscular blockers:** enhanced block.
- **Erythromycin:** antagonism.
- **Strong CYP3A4 inducers/inhibitors:** alter levels.

NURSING MANAGEMENT

- Give oral capsules with water, patient upright.
- Dilute IV and infuse slowly.
- Monitor stool frequency and character; stop and report diarrhoea.
- Monitor LFTs, CBC in long courses.
- Assess wound/infection response.

PATIENT EDUCATION

- Stop and seek advice if diarrhoea starts, even weeks after finishing.
- Swallow capsules with a full glass of water.
- Complete the course.

KEY POINTS

- Lincosamide — Gram-positive + anaerobes.
- Suppresses toxins in necrotising fasciitis/TSS.
- Highest risk of C. difficile colitis.
- Never IV bolus.

SOURCES CHECKED **[187]** Clindamycin 150 mg/ml Solution for Injection (UK SmPC) **[188]** Cleocin Phosphate, IM, IV (US label)

Full references p. 432–457